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Exercise at Altitude

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In Closing



On October 14, 2012, Austrian skydiver Felix Baumgartner set several world records for skydiving. On that date, Baumgartner piloted a helium balloon into the stratosphere at an altitude of 39 km (128,000 ft) (a balloon ascent record) and, leaving the balloon, jumped to Earth from that height (a record for parachute jumping from the highest altitude). In doing so, he became the first human to break the sound barrier outside a vehicle, reaching a velocity of 1,343 km/h (833 mph) during a 4 min and 19 s free fall.

At such altitudes, without a pressurized suit and breathing equipment, Baumgartner's body would have been unable to supply his tissues with oxygen, and he would have quickly lost consciousness. This specialized suit was also designed to protect him from the low ambient pressure, extreme cold (-52°C , or -61°F), friction, and other dangers. Had the suit failed, above 19.2 km he would have experienced the fatal condition known as ebullism, the formation of gas bubbles in his bodily fluids.

Instead, he deployed his parachute and landed safely in the New Mexico desert, illustrating not only an exceptional feat of human performance but also the power of technology in overcoming—for a brief period of time—the dangers of high altitude.

Our previous discussions of the physiological responses to exercise have all been based on the conditions that exist at or near sea level, where the **barometric pressure (P_b)** averages about 760 mmHg. Recall from [chapter 7](#) that barometric pressure is a measure of the total pressure that all of the gases composing the atmosphere exert on the body (and everything else). Regardless of the P_b , oxygen molecules always make up 20.93% of the air. The **partial pressure of oxygen (PO_2)** is that portion of P_b exerted only by the oxygen molecules in the air. At sea level, PO_2 is therefore 0.2093 times 760 mmHg, or 159 mmHg. Partial pressure is an important concept in understanding altitude physiology, because it is primarily the low PO_2 at altitude that limits exercise performance. Although the human body tolerates small fluctuations in PO_2 , large variations pose problems. This is evident when mountain climbers ascend to high altitudes where significantly reduced PO_2 can substantially impair physical performance and can even jeopardize life.

The reduced barometric pressure at altitude is referred to as a **hypobaric** environment or simply hypobaria (low atmospheric pressure). The lower atmospheric pressure also means a lower PO_2 in the inspired air, which limits pulmonary diffusion of oxygen from the lungs and oxygen transport to the tissues. The low PO_2 in the air is termed **hypoxia** (low oxygen), while the resulting low PO_2 in the blood is called **hypoxemia**.

In this chapter, we examine the unique characteristics of hypobaric, hypoxic environments and how these conditions alter physiological responses at rest and during physical activity, training, and sport performance. We cover these changes upon acute ascent to altitude, the ways in which these responses change as humans acclimate to high altitude, and specialized training strategies used by athletes to improve performance at altitude. We also examine several specific health risks associated with hypoxic environments.

Environmental Conditions at Altitude

Clinical problems associated with altitude were reported as early as 400 BC. However, most of the early concerns about ascent to high altitudes focused on the cold conditions at altitude rather than the limitations imposed by low air pressure. The initial landmark discoveries that led to our current understanding of the reduced P_b and PO_2 at altitude can be credited primarily to four scientists spanning the 17th through the 19th centuries. Torricelli (ca. 1644) developed the mercury barometer, an instrument that permitted the accurate measurement of atmospheric pressure. Only a few years later (1648), Pascal demonstrated a reduction in barometric pressure at high altitudes. Nearly 130 years after that (1777), Lavoisier described oxygen and the other gases that contribute to the total barometric pressure. Finally, in 1801, John Dalton set forth the principle (called Dalton's law of partial pressures) stating that the total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of those individual gases.

The deleterious effects of high altitude on humans that are caused by low PO_2 (hypoxia) were recognized in the late 1800s. More recently, a team of scientists led by the late John Sutton performed an intricate series of laboratory studies in the hypobaric chamber at

the U.S. Army Institute of Environmental Medicine. These experiments, known collectively as *Operation Everest II*, have significantly added to our understanding of exercise at altitude.²³

Based on the effects of altitude on performance, the following definitions are useful:¹

- Near sea level [below 500 m (1,640 ft)]: There are no effects of altitude on well-being or exercise performance.
- Low altitude [500-2,000 m (1,640-6,560 ft)]: There are no effects on well-being but performance may be diminished, especially in athletes performing above 1,500 m (4,921 ft). These performance decrements may be overcome with acclimation.
- Moderate altitude [2,000-3,000 m (6,560-9,840 ft)]: Effects on well-being in unacclimated individuals and decreased maximal aerobic capacity and performance are likely. Optimal performance may or may not be restored with acclimation.
- High altitude [3,000-5,500 m (9,840-18,000 ft)]: There are adverse health effects (including acute mountain sickness, discussed later in this chapter) in a large percentage of individuals and significant performance decrements, even after full acclimation.
- Extreme altitude [above 5,500 m (~18,000 ft)]: Severe hypoxic effects are experienced. The highest permanent human settlements are at 5,200 to 5,800 m (17,000-19,000 ft).

For our discussion, the term *altitude* refers to elevations above 1,500 m (4,921 ft), because few negative physiological effects on exercise or sport performance are seen below that altitude.

While the major impact of altitude on exercise physiology is attributable to the low PO_2 that ultimately limits oxygen availability to the tissues, the atmosphere at altitude also differs in other ways from sea-level conditions.

Atmospheric Pressure at Altitude

At sea level, the air extending to the outermost reaches of the earth's atmosphere (approximately 38.6 km, or 24 mi) exerts a pressure equal to 760 mmHg. At the summit of Mount Everest, the highest point on Earth (8,848 m, or 29,029 ft), the pressure exerted by the air above is only about 250 mmHg. These and related altitude differences are depicted in [figure 13.1](#).

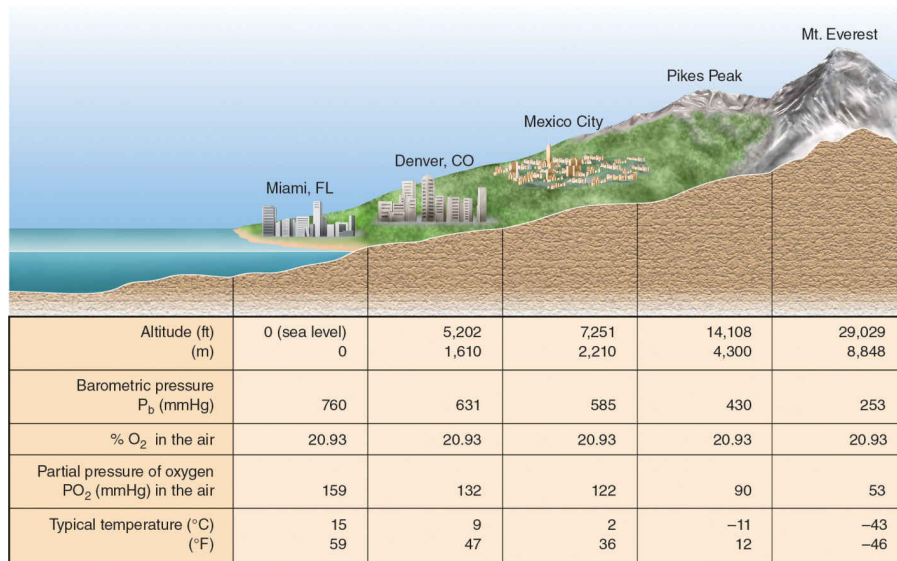


FIGURE 13.1 Differences in atmospheric conditions at sea level as altitude increases and barometric pressure falls accordingly. Note that the partial pressure of oxygen in the air decreases from 159 mmHg at sea level to only 53 mmHg at the summit of Mount Everest.

The barometric pressure on Earth does not remain constant. Rather, it varies somewhat with changes in climatic conditions, time of year, and the specific site at which the measurement is taken. On Mount Everest, for example, the mean barometric pressure varies from 243 mmHg in January to nearly 255 mmHg in June and July. These minor variations are of little interest at sea level, except to meteorologists because of their effect on weather patterns, but are of considerable physiological importance for a climber attempting to ascend Mount Everest without supplemental oxygen.

Although barometric pressure varies, the percentages of gases in the air that we breathe remain unchanged from sea level to high altitude. At any elevation, the air always contains 20.93% oxygen, 0.03% carbon dioxide, and 79.04% nitrogen. Only the partial pressures of these gases change. As shown in [figure 13.1](#), the

pressure that oxygen molecules in the air exert at various altitudes drops proportionally with decreases in the barometric pressure. The consequent changes in PO_2 have significant effects on the partial pressure of oxygen that reaches the lungs, as well as the gradients between the alveoli of the lungs and the blood (where oxygen is loaded) and between the blood and the tissues (where oxygen is unloaded). These effects are discussed in detail later in this chapter.

Air Temperature and Humidity at Altitude

Clearly, the low PO_2 at altitude has the greatest impact on exercise physiology. However, other environmental factors contribute to the ability to perform exercise as well. For example, air temperature decreases at a rate of about $1\text{ }^{\circ}\text{C}$ ($1.8\text{ }^{\circ}\text{F}$) for every 150 m (about 490 ft) of ascent. The average temperature near the summit of Mount Everest is estimated to be about $-40\text{ }^{\circ}\text{C}$ ($-40\text{ }^{\circ}\text{F}$), whereas at sea level the temperature would be about $15\text{ }^{\circ}\text{C}$ ($59\text{ }^{\circ}\text{F}$). The combination of low temperatures, low ambient water vapor pressure, and high winds at altitude poses a serious risk of cold-related disorders, such as hypothermia, frostbite, and nonfreezing cold injuries.

Water vapor has its own partial pressure, also known as the water vapor pressure (P_{H_2O}). Because of the cold temperatures at altitude, the water vapor pressure in the air is extremely low. Cold air holds very little water. Thus, even if air is fully saturated with water (100% relative humidity), the actual vapor pressure of water contained in the air is low. The extremely low P_{H_2O} at high altitude promotes evaporation of moisture from the skin (or clothing) surface, because of the high gradient between skin and air, and can lead quickly to dehydration. In addition, a large volume of water is lost through respiratory evaporation due to a combination of (a) a large vapor pressure gradient between warmed air leaving the mouth and nose and the dry air in the environment and (b) an increased respiration rate (discussed later) experienced at altitude.

Solar Radiation at Altitude

The intensity of solar radiation increases at high altitude for two reasons. First, at high altitudes, light travels through less of the atmosphere before reaching the earth. For this reason, less of the

sun's radiation, especially ultraviolet rays, is absorbed by the atmosphere at higher altitudes. Second, because atmospheric water normally absorbs a substantial amount of the sun's radiation, the low water vapor in the air at altitude also increases radiant exposure. Solar radiation may be further amplified by reflective light from snow, which is usually found at higher elevations.

In Review

- Altitude presents a hypobaric environment (one in which the atmospheric barometric pressure is reduced). Altitudes of 1,500 m (4,921 ft) or higher have a notable physiological impact on exercise performance.
- Although the percentages of the gases in the air that we breathe remain constant regardless of altitude, the partial pressure of each of these gases decreases with the decreased barometric pressure at higher altitudes.
- The low partial pressure of oxygen (PO_2) in the air at altitude is the environmental condition with the most profound physiological impact. Because the PO_2 in the lungs is low, PO_2 gradients between the alveoli of the lungs and the blood (where oxygen is loaded) and between the blood and the tissues (where oxygen is unloaded) are decreased.
- Air temperature typically decreases as altitude increases. Cold air can hold little water, so the air at altitude is dry. These two factors combine to increase susceptibility to cold-related disorders and dehydration.
- Because the atmosphere is thinner and drier at altitude, solar radiation is more intense at higher elevations. This effect is magnified when the ground is snow covered.

Physiological Responses to Acute Altitude Exposure

This section deals with how the human body responds to acute altitude exposure, emphasizing responses that can affect exercise and sport performance. The main concerns are respiratory, cardiovascular, and metabolic responses. Most physiological studies have been performed on healthy, fit young men; unfortunately, few studies on the effects of altitude have included women, children, or the elderly—populations whose responses to the conditions of altitude might differ from those described here.

Respiratory Responses to Altitude

Adequate oxygen supply to exercising muscles is essential to physical performance and, as seen in [chapter 8](#), depends on an adequate supply of oxygen being brought into the body, moved from the lungs to the blood, transported to the muscles, and adequately taken up into the muscles. A limitation in any of these steps can impair performance.

Pulmonary Ventilation

The sequence of steps leading to the transport of oxygen to working muscle begins with pulmonary ventilation, the active movement of gas molecules into the alveoli of the lungs (breathing). Ventilation increases within seconds of exposure to high altitude, both at rest and during exercise, because chemoreceptors in the aortic arch and carotid arteries are stimulated by the low PO_2 and signals are sent to the brain to increase breathing. The increased ventilation is primarily associated with an increased tidal volume and an even greater increase in respiratory rate. Over the next several hours and days, ventilation remains elevated to a level proportional to the altitude.

Increased ventilation acts much the same as hyperventilation at sea level. The amount of carbon dioxide in the alveoli is reduced. Carbon dioxide follows the pressure gradient, so more diffuses out of the blood, where its pressure is relatively high, and into the lungs to be exhaled. This “blowing off” of CO_2 causes blood PCO_2 to fall and blood pH to increase, a condition known as **respiratory alkalosis**. This alkalosis has two effects. First, it causes the oxyhemoglobin saturation curve to shift to the left (as discussed in the next section). Second, it helps keep the rise in ventilation caused by the hypoxic (low PO_2) drive from increasing even further. At a given submaximal exercise intensity, ventilation is higher at altitude than at sea level, but maximal exercise ventilation is similar.

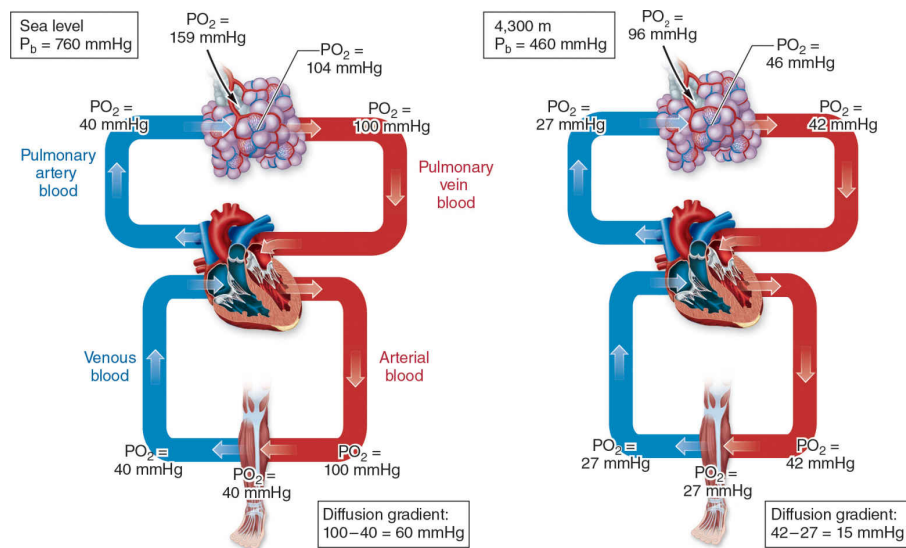


FIGURE 13.2 Comparison of the partial pressure of oxygen (PO_2) in the inspired air and in body tissues at sea level and at 4,300 m (14,108 ft) of altitude, the altitude of Pikes Peak, Colorado. As inspired PO_2 decreases, so does alveolar PO_2 . Arterial PO_2 is similar to that in the lungs, but the gradient for diffusion of O_2 into tissues, including muscle, is greatly reduced.



In an effort to offset respiratory alkalosis, the kidneys excrete more bicarbonate ion, the ions that buffer the carbonic acid formed from carbon dioxide. Thus, a reduction in bicarbonate ion concentration reduces the blood's buffering capacity. More acid remains in the blood, and the alkalosis is minimized.

Pulmonary Diffusion

Under resting conditions, pulmonary diffusion (diffusion of O_2 from the alveoli to the arterial blood) does not limit the exchange of gases between the alveoli and the blood. If gas exchange were limited or impaired at altitude, less oxygen would enter the blood, so the

arterial PO_2 would be much lower than the alveolar PO_2 . Instead, these two values are almost equal (figure 13.2). Therefore, the low arterial blood PO_2 , or hypoxemia, is a direct reflection of the low alveolar PO_2 and not a limitation of oxygen diffusion from the alveoli to the arterial blood.

As introduced in chapter 7, the oxygen cascade is a way of depicting the changes in PO_2 from the air through the tissues and into the venous circulation. Figure 13.3 shows the differences at various points in the oxygen cascade between sea level and an altitude of 5,800 m (19,029 ft).

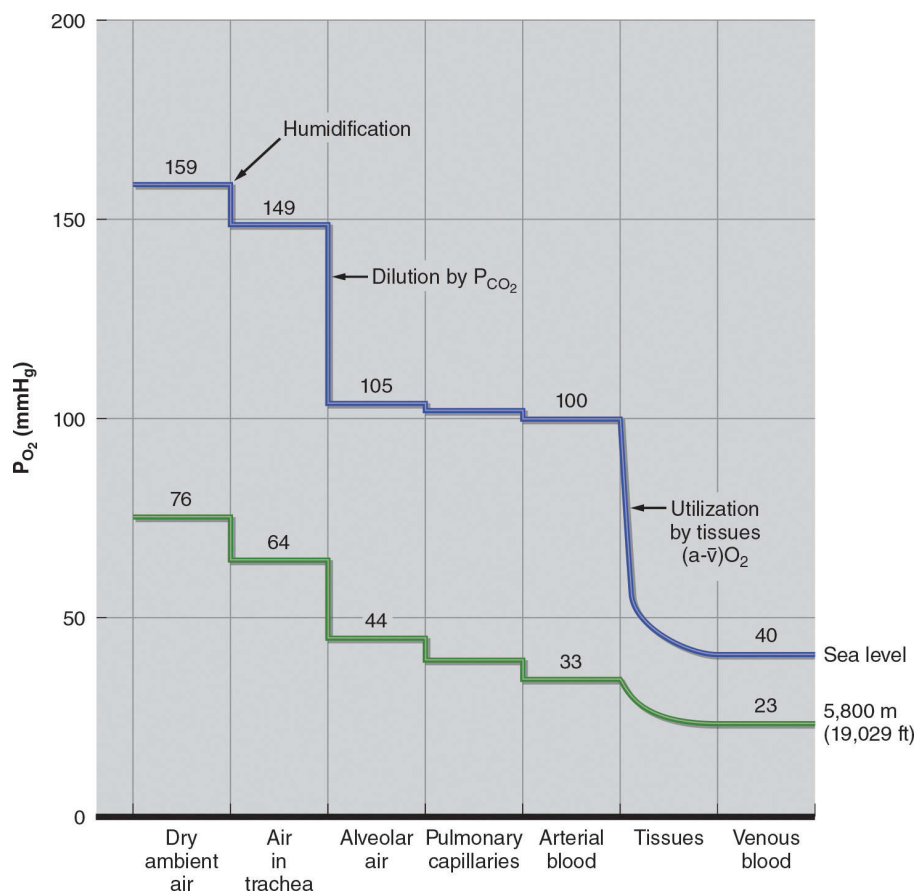


FIGURE 13.3 Comparison of the oxygen cascade at sea level and at an altitude of 5,800 m. Each step, from inspired air to tissue and venous blood shows a remarkable decrease in PO_2 , reducing the gradient for diffusion of O_2 into tissues, including muscle.



Oxygen Transport

As shown in [figures 13.2](#) and [13.3](#), the inspired PO_2 at sea level is 159 mmHg; however, it decreases to about 105 in the alveoli, primarily because of the addition of water vapor molecules ($P_{\text{H}_2\text{O}} = 47$ mmHg at 37 °C). When the alveolar PO_2 drops at altitude, fewer binding sites on the hemoglobin in the blood perfusing the lungs become saturated with O_2 . As depicted in [figure 13.4](#), the oxygen-binding (or oxyhemoglobin dissociation) curve for hemoglobin has a distinct S shape. At sea level, when alveolar PO_2 is about 104 mmHg, 96% to 97% of hemoglobin has O_2 bound to it. When PO_2 in the lungs is decreased to 46 mmHg at 4,300 m (14,108 ft), only about 80% of hemoglobin sites are saturated with O_2 . If the oxygen-loading portion of the curve were not relatively flat, far less O_2 would be taken up by the blood as it passes through the lungs, and binding would be extremely limited at altitude. Therefore, while arterial blood is still desaturated at altitude, the inherent shape of the oxyhemoglobin dissociation curve serves to minimize this problem.

A second adaptation occurs very early in altitude exposure that also aids in preventing the fall in arterial oxygen content. As mentioned earlier, a respiratory alkalosis accompanies the increased ventilation caused by acute altitude exposure. This increase in blood pH actually shifts the oxyhemoglobin dissociation curve to the left, as shown in [figure 13.3](#). The result is that, rather than 80% binding of oxygen to hemoglobin, 89% of hemoglobin is saturated with O_2 . Because of this shift, more oxygen binds to hemoglobin in the lungs and more oxygen is unloaded to the tissues at higher altitudes, where PO_2 is lower in both tissues.

Gas Exchange at the Muscles

[Figures 13.2](#) and [13.3](#) illustrate that arterial PO_2 at sea level is about 100 mmHg, and the PO_2 in body tissues is consistently about 40 mmHg at rest; so the difference, or the pressure gradient, between the arterial PO_2 and the tissue PO_2 at sea level is about 60 mmHg. However, when one moves to an elevation of 4,300 m (14,108 ft),

arterial PO_2 decreases to about 42 mmHg and the tissue PO_2 drops to 27 mmHg. Thus, the pressure gradient decreases from 60 mmHg at sea level to only 15 mmHg at the higher altitude. This represents a 75% reduction in the diffusion gradient. At 5,800 m (19,029 ft), the gradient shrinks to only 10 mmHg. Because the diffusion gradient is responsible for driving the oxygen from the hemoglobin in the blood into the tissues, this change in arterial PO_2 at altitude is a much greater consideration for exercise performance than the small reduction in hemoglobin saturation that occurs in the lungs.

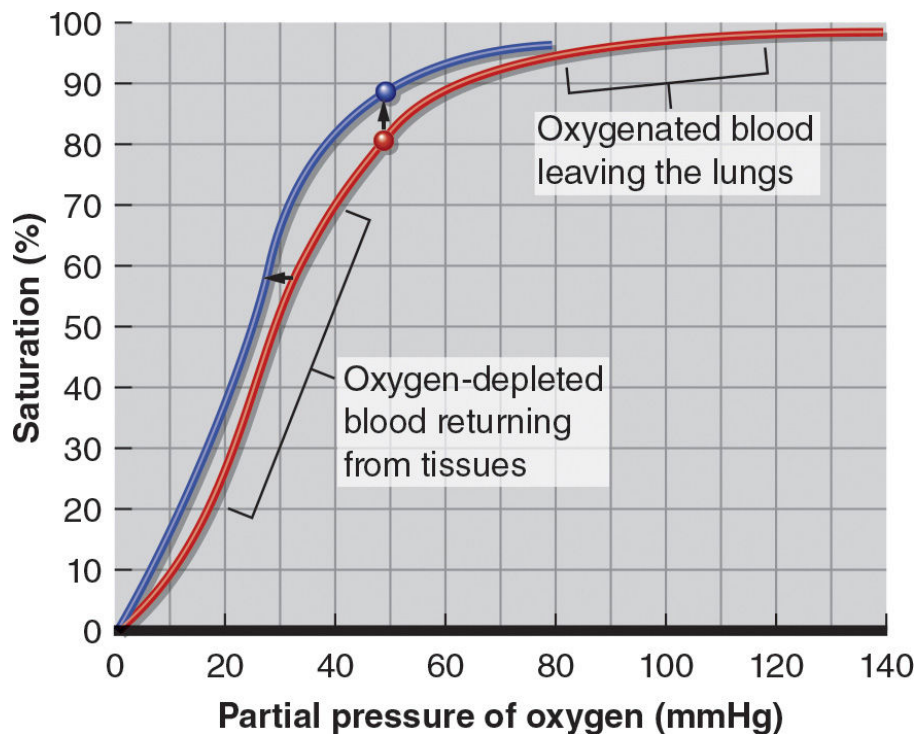


FIGURE 13.4 The S-shaped oxyhemoglobin dissociation curve at sea level (red line). When alveolar PO_2 is about 104 mmHg, 96% to 97% of hemoglobin is saturated with O_2 . The respiratory alkalosis with acute altitude exposure shifts the oxyhemoglobin dissociation curve leftward (blue line), partially compensating for the desaturation resulting from a drop in PO_2 .

Cardiovascular Responses to Altitude

As the respiratory system becomes increasingly limiting at altitude, the cardiovascular system likewise undergoes substantial changes to attempt to compensate for the decrease in arterial PO_2 that accompanies hypoxia.

Blood Volume

Within the first few hours of arrival at altitude, plasma volume begins to progressively decrease, and the decline reaches a plateau by the end of the first few weeks. This decrease in plasma volume is the result of both respiratory water loss and increased urine production. This combination of respiratory water loss and the excretion of fluid can reduce total plasma volume by up to 25%. Initially, the result of the plasma loss is an increase in the hematocrit, the percentage of the blood volume composed of red blood cells (containing hemoglobin). This adaptation—more red blood cells for a given blood flow—allows more oxygen to be delivered to the muscles for a given cardiac output. Over a period of weeks at altitude, this diminished plasma volume eventually returns to normal if adequate fluids are ingested.

Continued exposure to high altitude triggers the release of erythropoietin (EPO) from the kidneys; this is the hormone responsible for stimulating erythrocyte (red blood cell) production. This increases the total number of red blood cells and creates a greater total blood volume, which allows the person to partially compensate for the lower PO_2 experienced at altitude. However, this compensation is slow, taking weeks to months to fully restore red cell mass.

Cardiac Output

The preceding discussion clearly illustrates that the amount of oxygen delivered to the muscles by a given volume of blood is limited at altitude because of the reduced arterial PO_2 . A logical means to compensate for this is to increase the volume of blood delivered to the active muscles. At rest and during submaximal exercise, this is accomplished by increasing cardiac output. Since cardiac output is the product of stroke volume and heart rate, increasing either or both of these variables will increase cardiac output. Upon ascent to altitude, the sympathetic nervous system is stimulated, releasing norepinephrine and epinephrine, the main hormones that alter cardiac function. The increase in norepinephrine in particular persists for several days of acute altitude exposure.

When submaximal exercise is performed during the first few hours at altitude, stroke volume is decreased compared to sea-level values

(attributable to the reduced plasma volume). Fortunately, heart rate is increased disproportionately not only to compensate for the decrease in stroke volume but also to slightly increase cardiac output. However, this extra cardiac workload is not an efficient way to ensure sufficient oxygen delivery to the body's active tissues for prolonged periods. Consequently, after a few days at altitude, the muscles begin extracting more oxygen from the blood (increasing the arterial–venous oxygen difference), which reduces the demand for increased cardiac output, since $\dot{V}O_2 = \dot{Q} \times (a-\bar{v})O_2$ difference. The increase in heart rate and cardiac output peaks after about 6 to 10 days at high altitude, after which cardiac output and heart rate during a given exercise bout start to decrease.

At maximal or exhaustive work levels at higher altitudes, both maximal stroke volume and maximal heart rate are decreased, as is cardiac output. The decrease in stroke volume is directly related to the decrease in plasma volume. Maximal heart rate may be somewhat lower at high altitude as a consequence of a decrease in the response to sympathetic nervous system activity, possibly attributable to a reduction in β -receptors (receptors in the heart that respond to sympathetic nerve activation, thus increasing the heart rate). With a decreased diffusion gradient to move oxygen from the blood into the muscles, coupled with this reduction in maximal cardiac output, it is apparent why both $\dot{V}O_{2\max}$ and submaximal aerobic performance are hindered at altitude. In summary, hypobaric conditions significantly limit oxygen delivery to the muscles, reducing the capacity to perform high-intensity or prolonged aerobic activities.

Metabolic Responses to Altitude

Ascent to altitude increases the basal metabolic rate, possibly due to increases in both thyroxine and catecholamine concentrations. This increased metabolism must be balanced by an increased food intake to prevent body weight from decreasing, a common occurrence during the first few days at altitude because appetite declines as well. In individuals who maintain their body weight at altitude, there is an increased reliance on carbohydrate for fuel, both at rest and during submaximal exercise. Because glucose yields more energy than fat or protein per liter of oxygen, this adaptation is beneficial.

Table 13.1 summarizes the acute responses to altitude at rest and during submaximal exercise. Given the hypoxic conditions at altitude, and because any fixed amount of work at altitude represents a higher percentage of $\dot{V}O_{2\max}$, we would expect anaerobic metabolism to be increased. If this occurs, we would expect lactic acid production to increase at any given work rate above the lactate threshold. This is in fact the case upon arrival at altitude. However, with longer exposure to altitude, the lactate concentration in the muscles and venous blood at a given intensity of exercise (including maximal exertion) is lower, despite the fact that muscle $\dot{V}O_{2\max}$ does not change with adaptation to altitude. To date, there is no universally accepted explanation for this so-called lactate paradox.⁴

TABLE 13.1 Effects of Acute Hypoxia (Initial 48 h) on Physiological Responses at Rest and During Submaximal Exercise

System	Acute hypoxic effect at rest	Acute hypoxic effect at a given submaximal exercise intensity
Respiratory and oxygen transport	Immediate increase in ventilation (increased frequency > increased tidal volume) Decreased 2,3-diphosphoglycerate concentration Leftward shift in the oxyhemoglobin dissociation curve Stimulation of peripheral chemoreceptors Respiratory alkalosis	Increased ventilation
Cardiovascular	Decreased plasma volume Increased heart rate Decreased stroke volume Increased cardiac output Increased blood pressure	Increased heart rate Decreased stroke volume (due to decreased plasma volume) Increased cardiac output Increased $\dot{V}O_2$
Metabolic	Increased basal metabolic rate Decreased $(a-\dot{V})O_2$ difference	Greater utilization of carbohydrates for energy Increased lactate production initially, then lower Decreased blood pH
Renal	Diuresis Excretion of bicarbonate ions Increased release of erythropoietin	

Nutrition Needs at Altitude

In addition to the alterations in physiological systems and processes already described, several other considerations are important to note with ascent to altitude. At altitude, the body has a natural tendency to lose fluids through the skin (insensible water loss), the respiratory system, and the kidneys. This water loss is exaggerated with exercise as sweat evaporation increases from the wetted skin to the relatively dry air. These avenues of fluid loss dramatically increase

the risk of dehydration, and one should pay careful attention to staying hydrated. A rule of thumb at altitude is to consume at least 3 L of fluid per day; however, this must be tailored to individual needs. It may seem counterproductive to increase fluid intake when the decrease in plasma volume is taking place to help pack red blood cells. However, dehydration can negatively alter the body water balance among fluid compartments, so staying well hydrated and allowing the natural decrease in plasma volume to occur is sound advice.

Appetite decreases at altitude, and decreased food intake often accompanies that decline. This decreased energy consumption coupled with increased metabolic rates can lead to daily energy deficits of up to 500 kcal/day, resulting in weight loss over time. Consuming adequate calories to support exercise and recreational activities is important, and climbers should be instructed to eat more calories than their appetite suggests.

Finally, successful acclimation and acclimatization to high altitude depend on adequate iron stores in the body. Iron deficiency may prevent the increase in red blood cell production that occurs progressively over the first 4 weeks or so at altitude. Therefore, consumption of iron-rich foods and perhaps even iron supplements is recommended before and during altitude exposure.

In Review

- Altitude causes hypobaric hypoxia, resulting in a decreased partial pressure of oxygen in the inspired air, in the alveoli, in the blood, and at the tissue level.
- With acute exposure to altitude, a series of adaptations occur in an attempt to maximize oxygen delivery to the tissues. Pulmonary ventilation increases and pulmonary diffusion is reasonably well maintained, but oxygen transport is slightly impaired because hemoglobin saturation at altitude is reduced.
- Ventilation increases almost immediately upon exposure to hypoxia because the decreased PO_2 stimulates peripheral chemoreceptors. The increased rate and depth of breathing help offset even larger decreases in PO_2 in the body.
- The diffusion gradient that allows oxygen exchange between the blood and active tissue is substantially reduced at moderate and high altitudes; thus, oxygen uptake by muscle is impaired.

- A decrease in plasma volume initially increases red blood cell concentration, allowing more oxygen to be transported per unit of blood, partially compensating for the impaired oxygen binding to hemoglobin.
 - Upon initial ascent to altitude, cardiac output increases during submaximal work to compensate for the decreased oxygen content per liter of blood. It does so by increasing heart rate, because stroke volume falls with the fall in plasma volume.
 - During maximal work at altitude, stroke volume and heart rate are both lower, which reduces cardiac output. This reduced cardiac output, combined with the decreased pressure gradient, severely impairs oxygen delivery to tissues.
 - Ascent to altitude increases metabolic rate by increasing sympathetic nervous system activity. There is an increased reliance on carbohydrate for fuel, both at rest and during submaximal exercise.
 - The exaggerated fluid loss and general loss of appetite at altitude increase the risk of dehydration.
 - Decreased energy intake coupled with the increased energy expenditure of activity at altitude can lead to daily energy deficits and weight loss.
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Exercise and Sport Performance at Altitude

The difficulty of demanding physical exertion at high altitude has been described by many climbers. In 1925, E.G. Norton¹⁷ gave the following account of climbing without supplemental oxygen at 8,600 m (28,208 ft): “Our pace was wretched. My ambition was to do 20 consecutive paces uphill without a pause to rest and pant, elbow on bent knee, yet I never remember achieving it—13 was nearer the mark.” In this section we briefly consider how exercise and sport performance is affected by altitude.

Maximal Oxygen Uptake and Endurance Activity

Maximal oxygen uptake decreases as altitude increases (see [figure 13.5](#)). $\dot{V}O_{2\max}$ decreases little until the atmospheric PO_2 drops below 131 mmHg. This decline generally begins at an altitude of about 1,500 m (approximately 5,000 ft)—about the elevation of Denver, Colorado, and Albuquerque, New Mexico. At altitudes between 1,500 m and 5,000 m (5,000 and 16,400 ft), the decreased $\dot{V}O_{2\max}$ is due primarily to the reduced arterial PO_2 ; at higher elevations, a decreased maximal cardiac output further limits $\dot{V}O_{2\max}$. $\dot{V}O_{2\max}$

decreases approximately 8% to 11% for every 1,000 m increase (or 3% for every 1,000 ft increase) in altitude above 1,500 m. The rate of decline may become even steeper at very high altitudes (figure 13.6). Endurance athletes with a high $\dot{V}O_{2\max}$ at sea level have a competitive advantage at altitude if everything else is equal. As $\dot{V}O_{2\max}$ declines on arrival at altitude, competition at any given pace will be performed at a lower percentage of $\dot{V}O_{2\max}$.

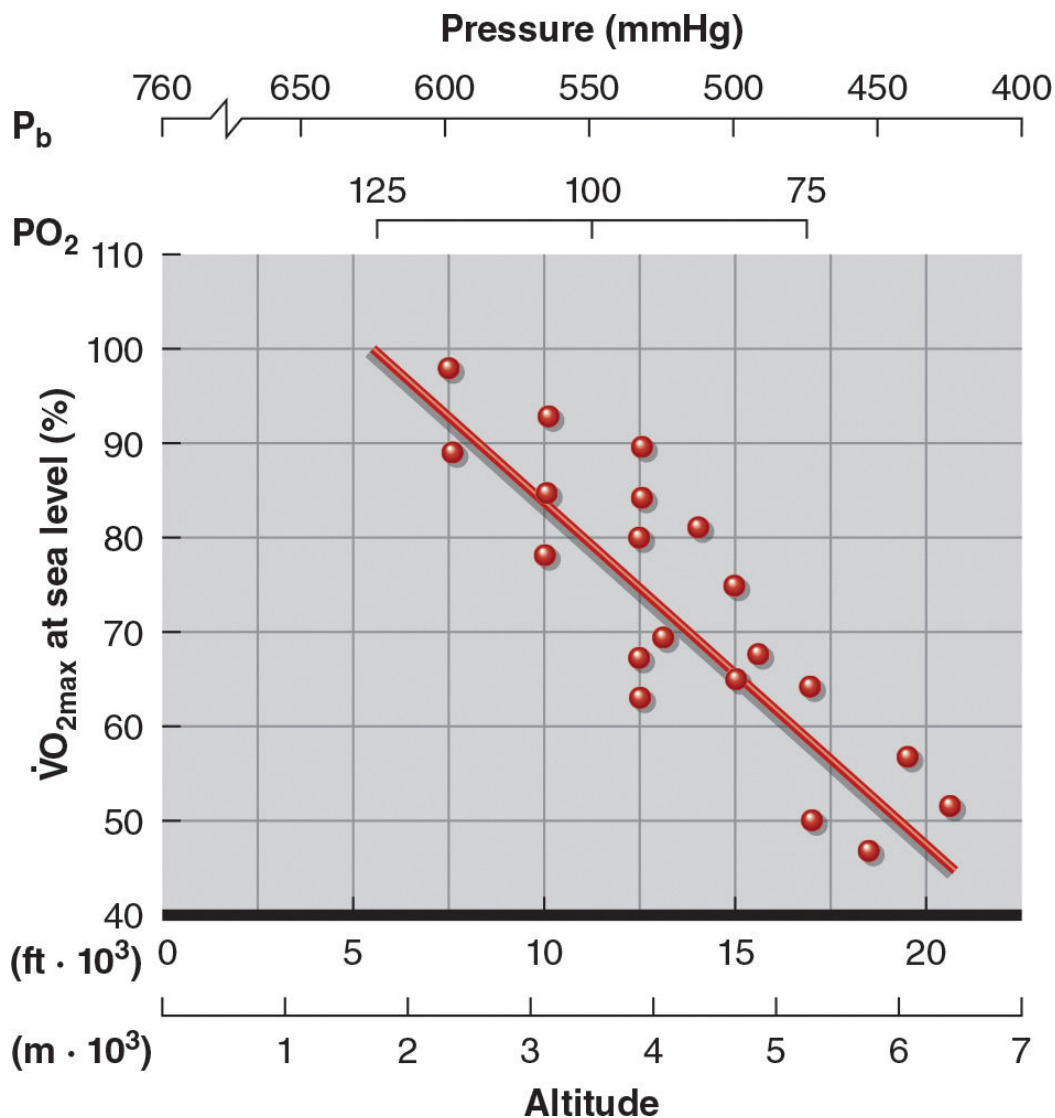


FIGURE 13.5 Changes in maximal oxygen uptake ($\dot{V}O_{2\max}$) with decreases in barometric pressure (P_b) and partial pressure of oxygen (PO_2). Values for $\dot{V}O_{2\max}$ are recorded as percentages of $\dot{V}O_{2\max}$ attained at sea level ($P_b = 760$ mmHg). Note that the decline in $\dot{V}O_{2\max}$ begins at about 1,500 m and is fairly linear. At the altitudes of Mexico City (2,240 m), Leadville, Colorado (3,180 m), and Nuñoa, Peru

(4,000 m), one's $\dot{V}O_{2\max}$ would be significantly below one's capacity at sea level or in Denver, Colorado (1,600 m).⁶
Data from Buskirk et al. (1967).

When men and women are matched for their initial aerobic fitness level, there are no sex differences in the rate of decline in $\dot{V}O_{2\max}$.

As shown in [figure 13.6](#), men climbing Mount Everest in a 1981 expedition experienced a change in $\dot{V}O_{2\max}$ from about 62 ml · kg⁻¹ · min⁻¹ at sea level to only about 15 ml · kg⁻¹ · min⁻¹ near the mountain's peak. Because resting oxygen requirements are about 3.5 ml · kg⁻¹ · min⁻¹, without supplemental oxygen these men had little capacity for physical effort at this elevation. A study by Pugh and coworkers,¹⁸ also shown in [figure 13.5](#), showed that men with $\dot{V}O_{2\max}$ values of 50 ml · kg⁻¹ · min⁻¹ at sea level would be unable to exercise, or even to move, near the peak of Mount Everest because their $\dot{V}O_{2\max}$ values at that altitude would decrease to 5 ml · kg⁻¹ · min⁻¹. Thus, most normal people with sea-level $\dot{V}O_{2\max}$ values below 50 ml · kg⁻¹ · min⁻¹ would not be able to survive without supplemental oxygen at the summit of Mount Everest because their $\dot{V}O_{2\max}$ values at such an altitude would be too low to sustain their body tissues. Enough oxygen would be consumed to barely meet their resting requirements.

Obviously, activities of long duration that place considerable demands on oxygen transport and uptake by the tissues are those that are most severely affected by the hypoxic conditions at altitude. At the summit of Mount Everest, $\dot{V}O_{2\max}$ is reduced to 10% to 25% of its sea-level value. This severely limits the body's exercise capacity. Because $\dot{V}O_{2\max}$ is reduced by a certain percentage, individuals with larger aerobic capacities can perform a standard work task with less perceived effort and with less cardiovascular and respiratory stress at altitude than those with a lower $\dot{V}O_{2\max}$. This may explain how noted mountaineers Reinhold Messner of Italy and Peter Habeler of Austria were able to reach the summit of Everest without supplemental oxygen in 1978—they obviously possessed high sea-level $\dot{V}O_{2\max}$ values.

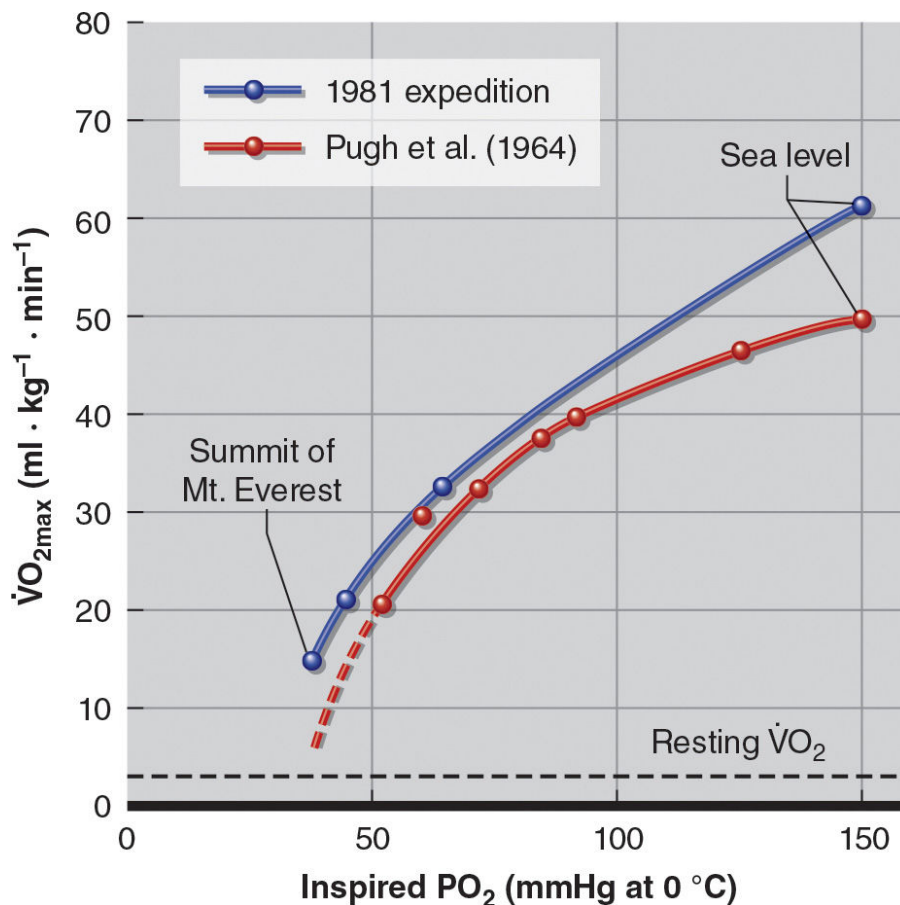


FIGURE 13.6 $\dot{V}O_{2\max}$ relative to the partial pressure of oxygen (PO_2) of the inspired air for two expeditions to Mount Everest.

Adapted by permission from J.B. West et al., "Maximal Exercise at Extreme Altitudes on Mount Everest," *Journal of Applied Physiology* 55 (1983): 688-698.

Anaerobic Sprinting, Jumping, and Throwing Activities

Whereas endurance events are impaired at altitude, anaerobic sprint activities that last less than a minute (such as 100 to 400 m track sprints) are generally not impaired by moderate altitude and can sometimes be improved. Such activities place minimal demands on the oxygen transport system and aerobic metabolism. Instead, most of the energy is provided through the adenosine triphosphate (ATP), phosphocreatine, and glycolytic systems.

In addition, the thinner air at altitude provides less aerodynamic resistance to athletes' movements. At the 1968 Olympic Games, for example, the thinner air of Mexico City clearly aided the performances of certain athletes. At Mexico City, world or Olympic records were set or tied in the men's 100 m, 200 m, 400 m, 800 m,

long jump, and triple jump events and in the women's 100 m, 200 m, 400 m, 800 m, 4 × 100 relay, and long jump events. Because similar results occurred in swimming events up to 800 m, some exercise scientists have questioned the role of lower air density in improved sprint performance. Interestingly, while shot put performance was not affected at the altitude of Mexico City, discus performance declined because there is less lift at low barometric pressures.

In Review

- Prolonged endurance performance suffers the most at high altitude because oxidative energy production is limited.
- Maximal oxygen consumption decreases in proportion to the decrease in atmospheric pressure, beginning to decline at about 1,500 m (4,921 ft).
- Anaerobic sprint activities that last 2 min or less are generally not impaired at moderate altitude. In some instances, sprint performance may be improved because the thinner air at altitude provides less resistance to movement.

Acclimation: Chronic Exposure to Altitude

When people are exposed to altitude over days, weeks, and months, their bodies gradually adjust to the lower oxygen partial pressure in the air. But, however well they acclimate to the conditions at high altitude, they never fully compensate for the hypoxia. Even endurance-trained athletes who live at altitude for years never attain the level of performance or the $\dot{V}O_{2\max}$ values that they might achieve at sea level. In this regard, acclimation to altitude is similar to heat acclimation discussed in [chapter 12](#). Heat acclimation improves performance and attenuates physiological strain during exercise in the heat compared to that experienced during the first few days; however, performance is still poorer than in cooler environments.

The following sections cover some of the physiological adaptations that occur with prolonged altitude exposure. These include changes at the pulmonary, cardiovascular, and muscle tissue (cellular) level. In general, these adaptations take longer to fully develop (several weeks to several months) than those associated with heat acclimation (typically 1-2 weeks). Generally, about 3 weeks

are needed for full acclimation to even moderate altitude. For each additional 600 m (1,970 ft) altitude increase, another week is needed on average. All of these beneficial effects are lost within a month of return to sea level. Many of these adjustments in resting and maximal exercise variables are shown in [figure 13.7](#).

Pulmonary Adaptations

One of the most important adaptations to altitude is an increase in pulmonary ventilation, both at rest and during exercise. Within 3 or 4 days at 4,000 m (13,123 ft), the increased resting ventilation rate levels off at a value about 40% higher than at sea level. Submaximal exercise ventilatory rate also plateaus at about 50% higher but over a longer time frame. Increases in ventilation during exercise remain elevated at altitude and are more pronounced at higher exercise intensities.

Blood Adaptations

During the first 2 weeks at altitude, the number of circulating erythrocytes increases. The lack of oxygen at altitude stimulates the renal release of erythropoietin or EPO. Within the first 3 h after arrival at a high elevation, the blood's EPO concentration increases; it then continues to increase for 2 or 3 days. Although blood EPO concentrations return to baseline levels in about a month, the **polycythemia** (increased red blood cells) may be evident for 3 months or more. After a person lives at 4,000 m (13,123 ft) for about 6 months, total blood volume (composed mainly of the red cell volume and the plasma volume) increases by about 10%, not only as a result of the altitude-induced stimulation of erythrocyte production but also because of plasma volume expansion.¹⁸

The percentage of total blood volume composed of erythrocytes is referred to as the hematocrit. Residents in the central Andes of Peru (4,540 m, or 14,895 ft) have an average hematocrit of 60% to 65% (properly termed an acclimatization rather than an acclimation response; see [chapter 12](#)). This is considerably higher than the average hematocrit of sea-level residents, which is only 45% to 48%. However, during 6 weeks of exposure to the Peruvian altitude, sea-level residents have shown remarkable increases in their hematocrit levels, up to an average of 59%.

As the volume of erythrocytes increases, so does the blood's hemoglobin (Hb) content (and concentration, after an initial decline; see [figure 13.7](#)). As noted in [figure 13.8](#), blood hemoglobin concentration tends to increase proportionately with increases in the elevation at which people reside. The data presented are for men. For women, however, the limited available data show a similar trend but with a lower concentration than for men at a given altitude. These adaptations improve oxygen-carrying capacity of a fixed volume of blood.

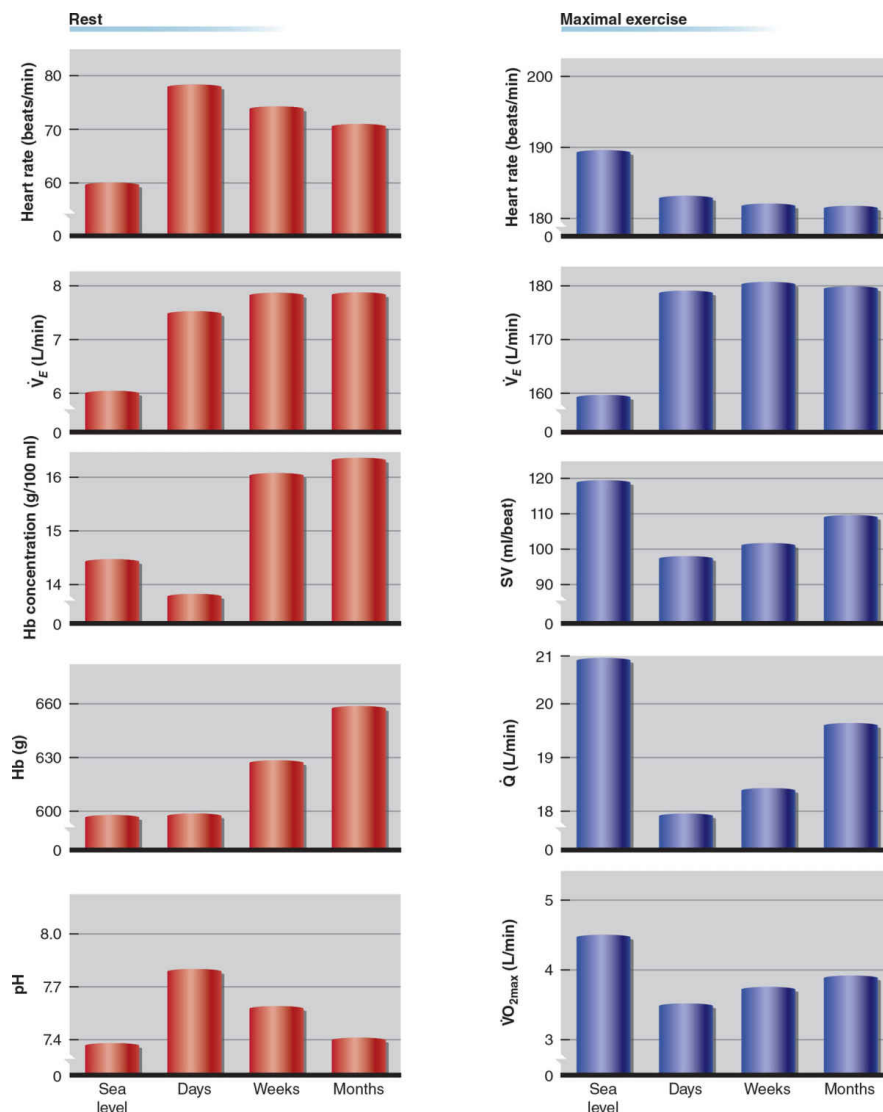


FIGURE 13.7 Physiological variables measured at sea level, after 2 or 3 days at altitude, and after weeks and months at altitude (3,000-3,500 m, or 9,843-11,483 ft). Both resting (left) and maximal exercise (right) variables are shown.

Drawn from data presented in Bartsch and Saltin (2008).



The reduction in plasma volume during acute altitude exposure reduces total blood volume, thus reducing submaximal and maximal cardiac output. But with acclimatization, as plasma volume increases over several weeks at altitude and as red blood cells continue to increase, maximal cardiac output increases. However, it does not return to sea-level values, as shown in [figure 13.7](#). Thus, overall oxygen delivery capacity is increased with acclimatization but not to the extent needed to achieve sea-level $\dot{V}O_{2\max}$ values.

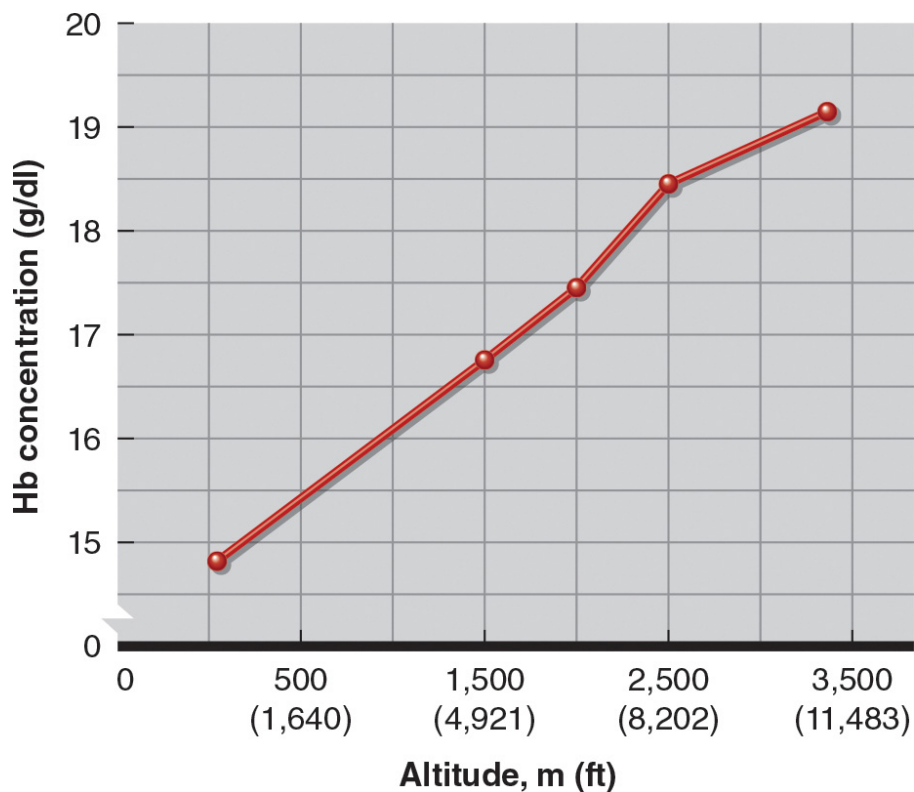


FIGURE 13.8 Hemoglobin (Hb) concentrations of men living at, and acclimatized to, various altitudes.

There is some debate about whether acclimation alters oxygen transport in the blood by changing the shape and position of the oxyhemoglobin dissociation curve ([figure 13.4](#)). The concentration of

2,3-diphosphoglycerate (2,3-DPG) increases in red blood cells, which shifts the curve to the right. This would favor the unloading of oxygen at the tissues (because more oxygen would be unloading from hemoglobin at any given low arterial PO_2), but this effect opposes the loading benefit of the respiratory alkalosis, a leftward shift. The net effect of both mechanisms is variable.

Muscle Adaptations

Although few attempts have been made to study muscle changes that occur during exposure to altitude, sufficient muscle biopsy data exist to indicate that muscles undergo significant structural and metabolic changes during ascent to altitude. In a study of climbers experiencing 4 to 6 weeks of chronic hypoxia on expeditions, muscle fiber cross-sectional area decreased, thus decreasing total muscle area. Capillary density in the muscles increased, which allowed more blood and oxygen to be delivered to the muscle fibers. Muscles' inability to meet exercise demands at high altitude might be related to a decrease in their mass and their ability to generate ATP.

The cause of the decreased muscle cross-sectional area within the first days and weeks at altitude is not fully understood. As previously discussed, prolonged exposure to high altitude frequently causes a loss of appetite and a noticeable weight loss. During a 1992 expedition to climb Mount McKinley, six men experienced an average weight loss of 6 kg, or 13 lb (D.L. Costill et al., unpublished data). Although part of this loss represents a general decrease in body weight and extracellular water, all of the men experienced a noticeable decrease in muscle mass. It seems logical to assume that much of this decrease in muscle mass is associated with loss of appetite and a wasting of muscle protein. Perhaps future studies on nutrition and body composition in mountain climbers will provide a more thorough explanation of the incapacitating influences of high altitude on muscle structure and function.

Several weeks at altitudes above 2,500 m (8,202 ft) reduce the metabolic potential of muscle, although this may not occur at lower elevations. Both mitochondrial function and glycolytic enzyme activities of the leg muscles (vastus lateralis and gastrocnemius) are significantly reduced after 4 weeks at altitude. This suggests that, in

addition to receiving less oxygen, muscles lose some of their capacity to perform oxidative phosphorylation and generate ATP. Unfortunately, no muscle biopsy data have been obtained from long-term residents at high altitudes to determine whether those individuals experience any muscular adaptations as a consequence of lifelong residence at these elevations.

Cardiovascular Adaptations

Studies conducted on endurance-trained runners in the late 1960s indicated that the decrement in $\dot{V}O_{2\max}$ when they first reached high altitude improved little for the duration of their exposure to hypoxia. Aerobic capacity remained unchanged for up to 2 months at altitude.⁶ Although the runners who previously had been exposed to altitude were more tolerant of hypoxia, their $\dot{V}O_{2\max}$ values and running performance were not significantly improved with acclimation. Because of the many adaptations that occur during acclimation to altitude, this lack of improvement in aerobic capacity and performance was unexpected. Perhaps these trained subjects had already attained maximal training adaptations and were unable to further adapt in response to altitude exposure. Or perhaps the reduced PO_2 of altitude made it more difficult for them to train at the same intensity and volume as at sea level. Both possibilities have some merit based on the available literature. While a ceiling effect on performance is more difficult to prove, athletes who train at altitude routinely have a difficult time maintaining their sea-level training intensities or volumes.

In Review

- Hypoxic conditions stimulate the renal release of EPO, which increases erythrocyte (red blood cell) production in bone marrow. More red blood cells means more hemoglobin. Although plasma volume decreases initially, which helps concentrate the hemoglobin, it eventually returns to normal. Normal plasma volume plus additional red blood cells increases total blood volume. All these changes increase the blood's oxygen-carrying capacity.
- Total muscle mass decreases after a few weeks at altitude, as does total body weight. Part of this decrease is from dehydration and appetite suppression. However, there is also protein breakdown in the muscles.

- Other muscle adaptations include decreased fiber area, increased capillary supply, and decreased oxidative enzyme activities.
 - While work capacity improves with altitude acclimation, the decrease in $\dot{V}O_{2\max}$ with initial exposure to altitude does not improve much over several weeks of exposure and typically never returns to sea-level values.
-

RESEARCH PERSPECTIVE 13.1

Human Adaptation to High Altitude: Tibetan and Sherpa Physiology

Three main regions of the world host human populations at an altitude over 4,000 m. These are the Tibetan plateau and Himalayan valleys, the South American Andes, and the Ethiopian Highlands. Of these regions, the Tibetan plateau is both the largest and the highest, and the human population there has successfully lived and reproduced at this high altitude for hundreds of generations. With hypoxia as a constant evolutionary pressure, these populations have adapted to life at high altitude, which is evident in the physiology of the Tibetan people who live at these altitudes compared to lowlanders.¹³

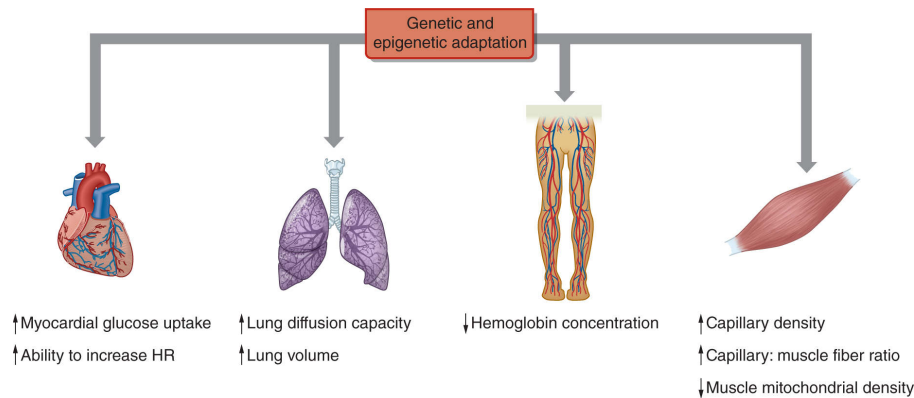
Since the early expeditions to the summit of Mount Everest, tales of Sherpas' extraordinary tolerance and exercise capacity at high altitude have provided anecdotal evidence of their physiological adaptations to the hypoxic environment. More recently, research studies aimed at exploring these unique populations have characterized a number of physiological adaptations that increase oxygen absorption and delivery in the face of hypoxia. Matched for age, height, weight, and smoking history, Tibetans demonstrate larger chest circumferences, total lung capacity, vital capacity, residual volumes, and tidal volumes compared to lowlanders while Sherpas also demonstrate a greater expiratory flow rates and forced vital capacity. These morphological and mechanical respiratory adaptations allow for greater respiratory volumes and improved lung diffusion capacity. Aiding in this improved diffusion capacity at the lung is a blunted hypoxic pulmonary vasoconstriction in Tibetan people. Hypoxic pulmonary vasoconstriction aids in ventilation and perfusion matching in healthy lowlanders but leads to pulmonary hypertension in lowlanders exposed to hypoxic conditions. This response is blunted in Tibetans, resulting in greater pulmonary perfusion for oxygen absorption at the alveoli. Interestingly, Tibetans do not have higher hemoglobin concentrations compared to lowlanders; in fact, they have a different set point for hypoxia-induced erythropoiesis and a blunted rise in hemoglobin with altitude.

At altitude, exercising Tibetans and Sherpas are able to generate greater maximal heart rates and increased cardiac outputs and to maintain stroke

volume when compared to lowlanders. This enhanced cardiac capacity likely facilitates superior work at altitude and does not induce right ventricular hypertrophy, as would be expected in lowlanders exposed to altitude for a prolonged period. This preservation of cardiac function despite chronic exposure to hypoxia may be due to metabolic adaptations that drive a preference for glucose as a substrate for the myocardium. Under resting conditions, the preferred substrate for cardiac muscle is normally fatty acids. However, Sherpas have an elevated myocardial glucose uptake compared to lowlanders under the same conditions. This switch toward glucose as a preferred fuel makes sense under hypoxic conditions, given that the ATP yield per oxygen molecule is much higher with glucose than with fatty acids.

Sherpas possess a greater number of capillaries and a lower muscle fiber cross-sectional area, which increases their ratio of capillaries to muscle fibers in skeletal muscle. This serves to increase convective and diffusive blood flow to deliver oxygen to the working muscle. Despite a 25% lower muscle mitochondrial volume density, Tibetans demonstrate a larger ratio of maximal oxygen consumption to mitochondrial volume. Similar to the myocardium, this seems to be achieved through an adaptive preference for glucose metabolism in the skeletal myocytes. Tibetans also have elevated myoglobin and antioxidant proteins, which both contribute to increased oxygen inflow and consumption under hypoxic conditions. Finally, Sherpas and Tibetans have a slight dominance of type I muscle fibers. This increased prevalence may explain the lactate paradox, in which people acclimatized to higher altitudes show lower than expected rise in blood lactate for a given work rate. Tibetan's predominance of slow type I muscle fibers seems to favor improved coupling between ATP demand and ADP supply, limiting lactate accumulation.

Overall, the unique physiology of Tibetan and Sherpa populations provides insight into the successful adaptation to hypobaric and hypoxic conditions. More recently, advancements in technological and analytical research methods have opened the door for the identification of the genetic and molecular mechanisms that contribute to these phenotypes¹⁴ (see figure). While these studies are in their infancy, it has been proposed that epigenetic changes (changes in heritable traits that are not explained by changes in the DNA sequence but are instead modulated by external environmental factors) may explain the ability of populations living at high altitude to rapidly acquire these adaptive features. Further studies in this area will continue to provide insight into the cellular and molecular mechanisms underlying human adaptive potential and will not only clarify the physiological adaptations of the Tibetans, but may also hold the key to further understanding in biomedical research and the study of human evolutionary theory.



Genetic and epigenetic adaptations lead to key physiological differences between Sherpas and Tibetans and lowlanders.

Altitude: Optimizing Training and Performance

We have considered the major changes that occur as the human body becomes acclimated to altitude and how these adaptations affect performance at altitude. But is there any advantage to training at altitude to improve performance at sea level? Are there advantages to training at altitude versus training at sea level when one must compete at altitude? And what about the relatively new concept of “live high, train low” to optimize performance?

Does Altitude Training Improve Sea-Level Performance?

Athletes have hypothesized for decades that training under hypoxic conditions, for example in an altitude chamber or by simply breathing low-oxygen gas mixtures, can improve sea-level endurance performance. Since many of the beneficial changes associated with altitude acclimation are similar to those conferred by aerobic training, can combining the two be even more beneficial? Can altitude training improve sea-level performance?

A strong theoretical argument can be made for altitude training. First, altitude training evokes substantial tissue hypoxia (reduced oxygen supply). This is thought to be essential for initiating the conditioning response. Second, the altitude-induced increase in red blood cell mass and hemoglobin content improves oxygen delivery on return to sea level. Although evidence suggests that these latter

changes are transient, lasting only several days, in theory this still should provide an advantage for the athlete.

Studying athletes at altitude poses additional problems because they are often unable to train at the same volume and intensity of effort as when at sea level. This was demonstrated in a group of elite female cyclists who performed self-selected maximum power outputs during high-intensity interval training. They completed trials under the following conditions: breathing atmospheric air (normoxia) and breathing a hypoxic gas mixture simulating 2,100 m (6,888 ft). The athletes' sustained (10 min) and short-term (15 s) power outputs at maximal intensity were reduced under hypoxic conditions.⁵ Training at even higher elevations, where acclimatization effects would be even more beneficial, causes even greater disruptions in training.

In addition, living and training at moderate to high altitude often causes athletes to dehydrate and to lose blood volume and muscle mass. These and other side effects tend to diminish the athletes' fitness and their motivation and tolerance for intense training. As a result, studies are difficult to interpret, but the value of altitude training for optimal sea-level performance has not been validated.

Live High, Train Low

When living and training at altitude, athletes are faced with the problem that the intensity of training is reduced because aerobic capacity and cardiorespiratory function are reduced at altitude. Thus, although athletes gain certain physiological benefits from being at altitude, they lose training adaptations associated with higher intensities of training. One way to get around this problem is to have athletes *live* at moderate altitude but *train* at low altitude, where training intensity is not compromised.

Researchers at the Institute for Exercise and Environmental Medicine in Dallas, Texas, conducted a series of studies in the mid-1990s to investigate altitude training for enhancing endurance performance. In one study,¹⁵ researchers divided 39 competitive runners into three equal groups: One (the high–low group) lived at moderate altitude (2,500 m, or 8,202 ft) and trained at low altitude (1,250 m, or 4,101 ft); one group (high–high) lived and trained at moderate altitude (2,500 m); and one group (low–low) lived and

trained at low altitude (150 m, or 490 ft). Using a 5,000 m time trial as the primary performance outcome measure, the researchers found that the high–low group was the only group to significantly improve their running performance, even though both the high–low and high–high groups increased their $\dot{V}O_{2\max}$ values by 5% in direct proportion to their increase in red cell mass. Thus, there appear to be performance benefits from living at moderate altitude but going to lower elevations to maximize training intensity.

VIDEO 13.1 Presents Ben Levine on the physiology behind the live high–train low approach.



This was tested more recently by the same scientists working with a group of 14 elite male and eight elite female runners, with all but two ranked in the U.S. top 50 for their event. These athletes lived at 2,500 m (8,202 ft) and trained at 1,250 m (4,101 ft) over a period of 27 days. Testing was conducted at sea level both the week before and the week after the 27 days of living at altitude. Sea-level 3,000 m time trial performance increased by 1.1% and $\dot{V}O_{2\max}$ increased by 3.2% as a result of this intervention.²¹ [Figure 13.9](#) illustrates the difference in race time performance for both studies, with values expressed as the percentage change before and after altitude exposure. These differences are plotted by the prealtitude race time, expressed as a percentage of the existing U.S. record in the event at the time of the time trial.

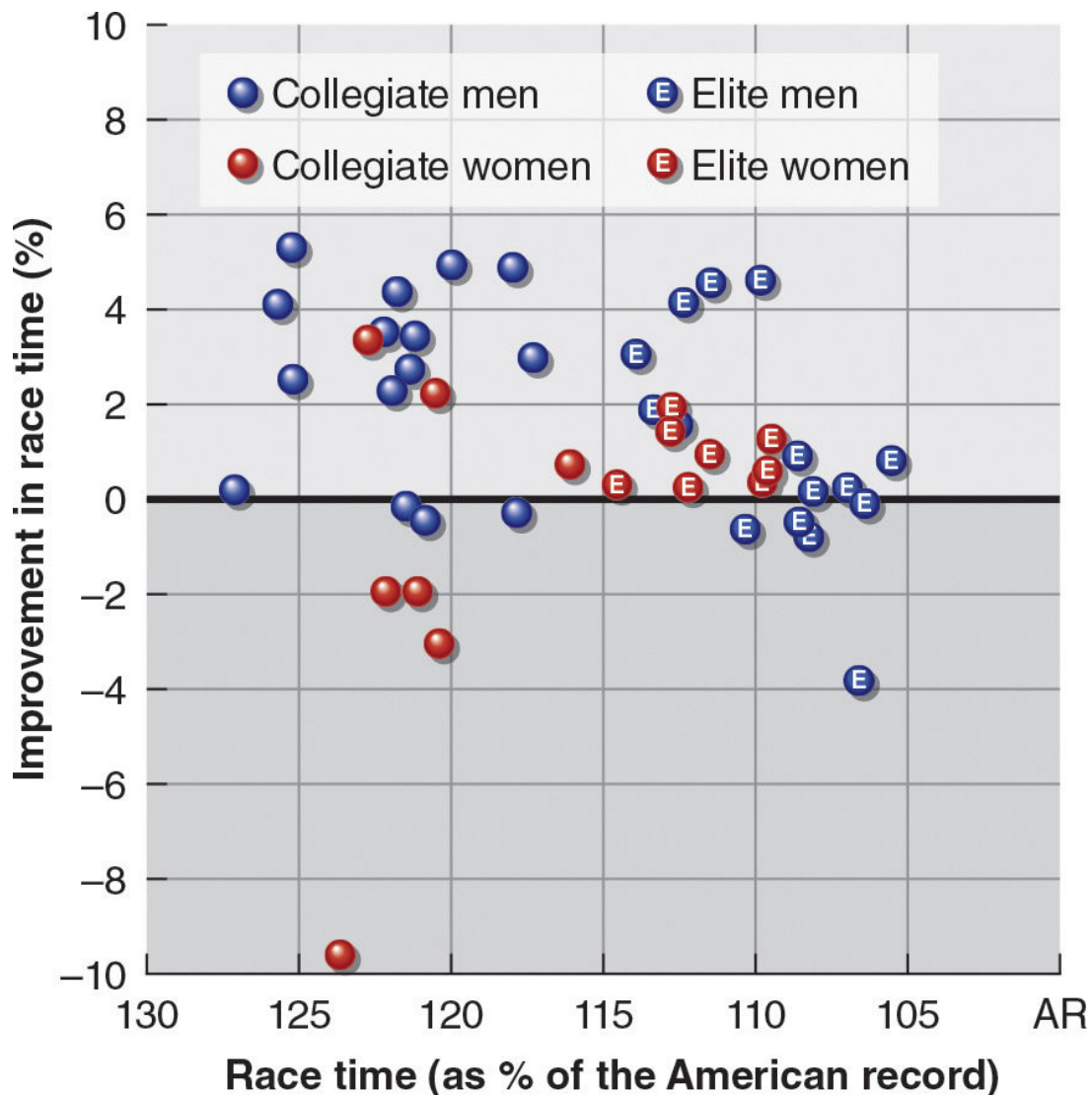


FIGURE 13.9 Improvement in race time (%) in elite male and female runners²¹ and collegiate male and female runners¹⁵ following 4 weeks of living at altitude but training at 1,250 m (4,100 ft). See the text for details.

Multiple studies have now shown a benefit of living and sleeping at high altitudes but training at low altitude for increasing sea-level $\dot{V}O_{2\max}$ or improving sea-level aerobic performance in elite endurance athletes. These improvements have been linked to an increase in the oxygen-carrying capacity of the blood. Chronic living above 2,500 m (8,202 ft) induces such hematological acclimatization features in most athletes, although there is some variability. A recent study examined whether there is a minimum threshold “living” altitude for improvements in sea-level performance.⁸ That is, do athletes who

live at relatively higher altitudes demonstrate greater performance improvements in a live high–train low setting?

In the study, 48 trained distance runners were randomly assigned to live at one of four altitudes: 1,780 m (5,840 ft), 2,085 m (6,841 ft), 2,454 m (8,051 ft), or 2,800 m (9,186 ft). The athletes all trained together at altitudes between 1,250 m (4,101 ft) and 3,000 m (9,842 ft). Upon return from the live high–train low altitude camp, red blood cell mass and EPO concentrations increased similarly in all four groups, but EPO returned to baseline sooner in the 1,780-m group. Sea-level $\dot{V}O_{2\max}$ was improved for all groups as expected (figure 13.10). However, performance on a 3 km (1.9 mi) run at sea level was significantly improved only in the groups that had lived at the middle two altitudes. It appears that increased red blood cell mass is necessary, but not sufficient, for improved performance. Also, there appears to be an optimal living altitude to optimize performance gains in such a setting.

It appears that team sports performance can likewise be enhanced using the live high–train low strategy. In a study of 32 elite male field hockey players,³ 14 days of living high (2,800–3,000 m) and training low as a team increased hemoglobin mass by 3% and shuttle-run performance by 21%. Moreover, these effects lasted for at least 3 weeks posttraining.

Optimizing Performance at Altitude

What can athletes who normally train at sea level but must compete at altitude do to prepare most effectively for competition? Although not all combinations have been attempted and research thus far is not conclusive, it appears that athletes have two viable options. One option is to compete as soon as possible after arriving at altitude, that is, within 24 h of arrival. This strategy does not provide the beneficial effects of acclimation, but the altitude exposure is brief enough that the classic symptoms of altitude sickness are not yet totally manifest. After the first 24 h, the athlete's physical condition often worsens because of the untoward effects of acute altitude exposure, such as dehydration, headache, and sleep disturbances. However, one laboratory study that asked whether there was a performance advantage to competing only 2 h after arrival at altitude

showed minimal differences in cycling performance and no physiological changes that would provide an advantage¹¹ compared to a 14 h exposure to the simulated altitude (2,500 m).

Another option is to train at higher altitudes for a minimum of 2 weeks before competing. But not even 2 weeks is sufficient for *total* acclimation, which would require a minimum of 3 weeks, and usually even longer. As previously mentioned, several weeks of intense aerobic training at sea level to increase the athletes' $\dot{V}O_{2\max}$ will allow them to compete at altitude at a lower relative intensity ($\% \dot{V}O_{2\max}$) than if they had not trained aerobically.

Extended training for optimal performance at altitude requires an elevation between 1,500 m (4,921 ft), which is considered the lowest level at which an effect will be noticed, and 3,000 m (9,842 ft), which is the highest level for efficient conditioning. Work capacity is reduced during the initial days at altitude. For this reason, when first reaching higher altitudes, athletes should reduce workout intensity to between 60% and 70% of sea-level intensity, gradually working up to full intensity within 14 days.

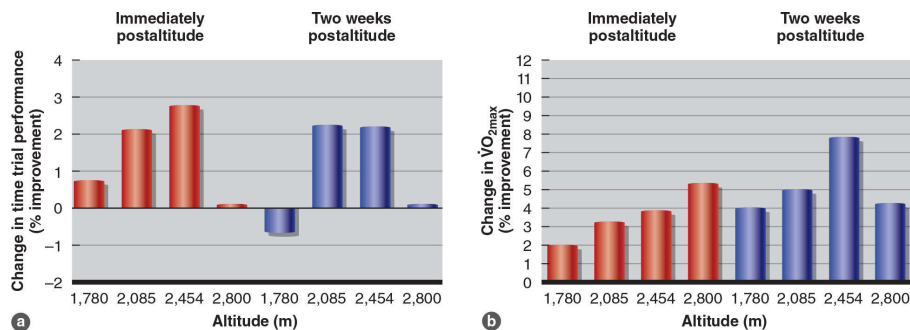


FIGURE 13.10 (a) Improvement in 3K time trial (%) and (b) $\dot{V}O_{2\max}$ immediately upon return to sea level and 2 weeks later in elite collegiate distance runners⁸ after 4 weeks of living at one of the four altitudes shown but training together as a group. See the text for details.

Adapted by permission from R.F. Chapman et al., "Defining the "Dose" of Altitude Training: How High to Live for Optimal Sea Level Performance Enhancement," *Journal of Applied Physiology* 116, no. 6 (2014): 595-603.

RESEARCH PERSPECTIVE 13.2

Altitude Training for Swimmers

The classical "live high, train high" approach to altitude training has been used by athletes since the late 1960s to improve performance at sea level. In a series of studies published in the 1990s, the "live high, train low" strategy

was shown to improve running performance in collegiate athletes by increasing red cell mass (the altitude adaptation) and maintaining high-intensity training velocities (possible only at low altitude). Since that time, altitude training has been adopted by endurance athletes of all disciplines, including elite swimmers. However, no studies have examined the effectiveness of altitude training for improved swim performance. Because of this disconnect between a lack of research evidence and common practical use of altitude training in swimmers, an international group of investigators recently conducted a multidisciplinary research project to study the effects of altitude training on swimmers' performance and health.

In this study (The Altitude Project), scientists examined the effects of four different altitude training paradigms on performance, $\dot{V}O_{2\max}$, oxygen kinetics, and total hemoglobin mass.²⁰ To do this, 61 swimmers were divided into four groups: (1) living and training at moderate altitude (2,320 m) for 4 weeks (Hi-Hi4), (2) living and training at moderate altitude for 3 weeks (Hi-Hi3), (3) living at altitude (2,320 m) and training at both moderate and low (690 m) altitude for 4 weeks (Hi-HiLo), and (4) living and training near sea level (190 or 655 m) for 4 weeks (Lo-Lo). Performance and physiological variables were tested before, once a week during, immediately after, and throughout a 4-week sea-level recovery from these training regimens. The researchers found that all four training regimens improved time trial performance by ~3.5% following 1 week of recovery, while Hi-HiLo achieved the greatest improvements in time trial performance compared to the other training regimens. Total hemoglobin mass increased in both Hi-Hi groups but did not increase in the Hi-HiLo or Lo-Lo groups. $\dot{V}O_{2\max}$ and oxygen kinetics were unchanged by any of the training regimens, a result that is likely explained by the elite status of these athletes and the limitations to improving $\dot{V}O_{2\max}$ beyond an already very high value.

Overall, the study findings led to the conclusion that the performance enhancement observed in the Hi-HiLo group was not linked to changes in $\dot{V}O_{2\max}$, hemoglobin mass, or oxygen kinetics, but that “living high, training high and low” for 4 weeks has the potential to improve swimming performance above altitude or sea-level training alone. The physiological alterations responsible for altitude acclimatization and subsequent training effects are complex, and a deeper understanding will require future investigations targeting the specific mechanisms and mediators.

Artificial Altitude Training

The largest and most important adaptations to altitude are physiological changes caused by the hypoxia experienced there, so we might anticipate that people could achieve similar adaptations simply by breathing gases with a low PO_2 . But no evidence supports

the idea that brief periods (1-2 h per day) of breathing hypoxic gases or hypobaric mixtures induce even a partial adaptation similar to that observed at altitude. Nor does such artificial hypoxic training confer any advantage for endurance performance at sea level.¹⁹ On the other hand, alternating periods (lasting between 5 and 14 days) of training at 2,300 m (7,546 ft) and at sea level adequately stimulated altitude acclimation in a group of elite middle-distance runners.⁹ Staying at sea level for up to 11 days did not interfere with the usual adjustments to altitude as long as training was maintained.

The favorable results of the studies on living high and training low have stimulated considerable interest in how this concept might be applied without having to send athletes to altitude to live. One approach has been to develop a hypoxic apartment where athletes sleep and live. The gas mixture inside the apartment is adjusted so that nitrogen represents a higher percentage of the inspired air, reducing the percentage of oxygen in the inspired air as well as its partial pressure. Pioneered by Finnish sport scientists, these apartments can simulate altitudes between 2,000 m (6,562 ft) and 3,000 m (9,842 ft) as the nitrogen and oxygen percentages in the inspired air are adjusted to reduce the partial pressure of oxygen to those levels associated with 2,000 to 3,000 m altitude. Hypoxic sleeping devices or tents have also been proposed.

Unfortunately, at this time, few carefully controlled scientific studies exist to confirm whether these apartments or sleeping devices actually improve performance and physiological function. One recent meta-analysis (a statistical approach that combines data from multiple studies to draw conclusions) reported that natural live high–train low approaches provide the best results for enhancing performance in elite athletes, while some nonelite exercisers seem to benefit from the artificial approaches.² However, the authors cautioned that improvements seen in these subelite athletes could have been due to a placebo effect. Ethical issues have also been raised about the use of such devices.

In Review

- Living at high altitudes and training at low altitudes currently appears to be the best practice for improving subsequent sea-level performance.

- Athletes who must perform at altitude should do so as early after arrival as possible, certainly within 24 h of arrival, before the detrimental side effects that occur at altitude become too great.
 - Alternatively, athletes who must perform at altitude could train at an altitude between 1,500 m (4,921 ft) and 3,000 m (9,842 ft) for a minimum of 2 weeks (more is better) before performing.
 - There is no evidence that brief periods (1-2 h per day) of breathing hypoxic gases or hypobaric gas mixtures induce even a partial adaptation similar to that observed at altitude.
-

Health Risks of Acute Exposure to Altitude

A large proportion of people who ascend to moderate and high altitudes experience symptoms of **acute altitude (mountain) sickness**. This disorder is characterized by symptoms such as headache, nausea, vomiting, dyspnea (difficult breathing), and insomnia. These symptoms can begin anywhere from 6 to 48 h after arrival at high altitude and are most severe on days 2 and 3. Although not life threatening, acute altitude sickness can be incapacitating for several days or longer. In some cases, the condition can worsen. The victim can develop the more lethal altitude-related illnesses of high-altitude pulmonary edema or high-altitude cerebral edema.

Acute Altitude (Mountain) Sickness

The incidence of acute altitude sickness varies with the altitude, the rate of ascent, and the individual's experience and susceptibility. Several studies have been conducted to determine the incidence of acute altitude sickness in groups of novice hikers (tourists) and more experienced climbers. Results vary widely, ranging from a frequency of less than 1% to almost 60% at altitudes of 3,000 to 5,500 m (9,842-18,045 ft) (figure 13.11). Forster,¹⁰ however, reported that 80% of those who ascended to the top of Mauna Kea (4,205 m, or 13,796 ft) on the island of Hawaii experienced some symptoms of acute altitude sickness. Another study showed that at elevations of 2,500 to 3,500 m (8,202-11,483 ft), altitudes commonly experienced by recreational skiers and hikers, the incidence of acute altitude

sickness was about 7% for men and 22% for women, but the reason for this sex difference is unclear.²²

Although the precise underlying cause of acute altitude sickness is not fully understood, it appears that people who experience the greatest distress also have a low ventilatory response to hypoxia. This inadequate ventilation allows PO_2 to decrease further and carbon dioxide to accumulate in the tissues, and these two factors may induce most of the symptoms associated with altitude sickness.

Headache is the most common symptom associated with ascent to high altitude. Headache is rarely experienced below 2,500 m (~8,000 ft), but ascent to 3,600 m (~12,000 ft) results in headache in the majority of people. The headache at altitude, which many sufferers describe as continuous and throbbing, is typically worse in the morning and after exercise. Alcohol consumption worsens the symptoms. The precise mechanism is unsubstantiated, but hypoxia causes dilation of the cerebral blood vessels, so stretching of pain receptors in these structures is a likely cause.

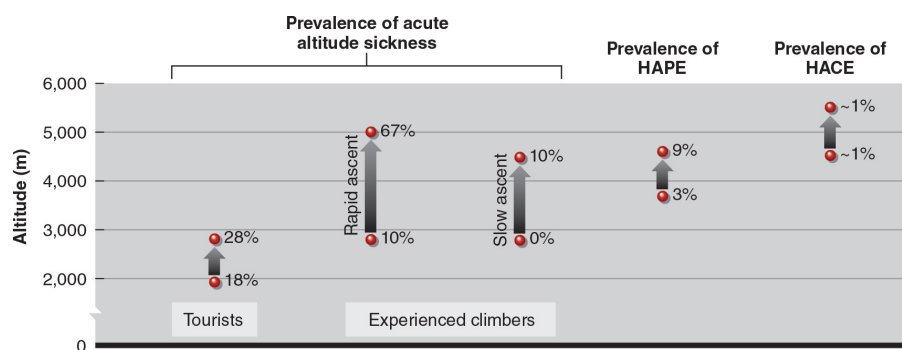


FIGURE 13.11 Change in the reported prevalence of acute altitude sickness, high-altitude pulmonary edema (HAPE), and high-altitude cerebral edema (HACE) as a function of altitude, experience, and, in the case of acute altitude sickness, rate of ascent.

Adapted from compiled data presented in Bartsch and Saltin (2008).

RESEARCH PERSPECTIVE 13.3

Should Athletes Live *Extra* High and Train Low?

Endurance exercise performance is impaired with acute altitude exposure, and athletes who normally live and train at sea level commonly plan to live and train for some period of time at altitude before competing in events scheduled to take place at altitude. This period of acclimatization reduces altitude-mediated declines in performance, but from a practical standpoint,

the athletes must consider both how long to arrive at altitude before the event and at what altitude they plan to reside relative to the altitude of the event. Research data suggest that athletes competing at altitude should arrive 14 days prior to competition. Performance improves significantly at altitude over the first 14 days with minimal improvements after that. However, the best altitude to reside at during that time remains unclear. Studies exploring the effects of different altitudes on performance at altitude have utilized the same acclimatization and performance altitudes or lower acclimatization altitudes for preacclimatization before high-altitude exposure. To date, only one study has explored whether living at a higher altitude than the competition altitude is more advantageous than the previously studied paradigms.

A recent study examined the decline in competitive distance running performance between sea level and low altitude (1,780 m) as a function of living altitudes and days of residence at altitude in 48 collegiate track and cross country athletes.⁷ The research team hypothesized that athletes living at higher altitudes than the altitude of the competition would have smaller declines in performance. To test this hypothesis, all subjects completed 4 weeks of sea-level training and were then randomized into one of four groups: live at 1,780 m, live at 2,085 m, live at 2,454 m, or live at 2,800 m. Each group was transported by car to their designated training altitude camp and lived and trained there for 28 days. Once a week, every training group performed a high-intensity training bout at low altitude (1,250 m). Subjects completed 3,000 m performance trials on the track at sea level, at baseline, 6 days before departure for altitude camp, and at 1,780 m on days 5, 12, 19, and 26 of the altitude camp. $\dot{V}O_{2\max}$ plasma volume, total blood volume, and red blood cell volume were measured once at sea level and once after the 28 days of altitude training.

All of the groups living at altitude had increases in red blood cell volume, but there were no differences between groups. $\dot{V}O_{2\max}$ increased in the two highest altitude groups (2,085 and 2,454 m) but did not improve in the 1,780 or 2,800 m groups. In terms of performance, athletes living at 2,454 and 2,800 m had a larger decline in performance compared to the 1,780 m group after 5 days at altitude. The 1,780 m group did not show changes in performance across the 26 days at altitude, while all of the groups living higher than the competition altitude showed improvements from day 5 to day 19 but no further improvement at day 26. The study team concluded that an endurance athlete competing acutely at low to moderate altitude should live at the altitude of the competition and not higher. For athletes living ~300 to 1,000 m higher than the competition altitude, acute altitude performance may be much worse and may require much longer (up to 19 days) acclimatization to minimize performance decrements.

Another consequence of acute altitude sickness is an inability to sleep even if the individual is markedly fatigued. Studies have shown that the inability to achieve satisfying sleep at altitude is associated with an interruption in the sleep stages. Additionally, some people suffer from a pattern of interrupted breathing, called **Cheyne-Stokes breathing**, which prevents them from falling asleep and remaining asleep. Cheyne-Stokes breathing is characterized by alternating periods of rapid breathing and slow, shallow breathing, usually including intermittent periods in which breathing completely stops. The incidence of this irregular breathing pattern increases with altitude, occurring in 24% of people at 2,440 m (8,005 ft), 40% of people at 4,270 m (14,009 ft), and almost everyone at altitudes above 6,300 m (20,669 ft).²⁴

The first few days at high altitude are associated with reduced exercise performance and increased risk of acute mountain sickness (AMS). How can exercisers and athletes prevent acute altitude sickness? Even athletes who are highly endurance trained before altitude exposure seem to have little protection against the effects of hypoxia, and it is difficult to determine which athletes may be susceptible to these symptoms unless this is suggested by a prior history of acute altitude sickness.

People can usually prevent acute altitude sickness by ascending to altitude gradually, spending periods of a few days at lower elevations. A gradual ascent of no more than 300 m (984 ft) per day at elevations above 3,000 m (9,842 ft) has been suggested to minimize the risks of altitude sickness. Scientists at the U.S. Army Research Institute for Environmental Medicine have conducted studies examining the efficacy of various staging and hypoxia treatments as preacclimatization strategies.^{12,16} To minimize symptoms associated with AMS and preserve performance capabilities at altitudes above 4,000 m (13,123 ft), evidence suggests the following:

- Ascending to moderate altitudes of >1,500 m for 1 to 2 days induces ventilatory acclimatization.
- Six days at 2,200 m substantially reduces AMS and improves exercise performance after rapid ascent to >4,000 m.

- Five or more days at 3,000 m within the last 2 months significantly reduces AMS.
- The longer the residence at moderate altitudes, the better the ventilatory adaptations and prevention of AMS at high altitude.
- Exercise training during these preexposures may improve physical performance even more.
- Strategies involving hypobaric hypoxia (actual altitude exposures, chambers, and tents) are much more effective than normobaric hypoxia (breathing low-oxygen mixtures at sea level).

Of the drugs that have been used to reduce symptoms in those who develop acute altitude sickness, acetazolamide started the day before ascent is the only established preventive measure. Acetazolamide is sometimes combined with a steroid such as dexamethasone. Both drugs must be used with medical supervision. Of course, the definitive treatment for severe acute mountain sickness is a retreat to lower altitude, but high-flow oxygen and the use of hyperbaric rescue bags are also effective in severe cases.

High-Altitude Pulmonary Edema

Unlike AMS, **high-altitude pulmonary edema (HAPE)**, which is the accumulation of fluids in the lungs, is life threatening. The cause of HAPE is likely related to the pulmonary vasoconstriction resulting from hypoxia, causing blood clots to form in the lungs. Remaining tissue becomes overperfused, and fluid and protein leak out of the capillaries. This seems to occur most frequently in unacclimatized people who rapidly ascend to altitudes above 2,500 m (8,202 ft). The disorder occurs in otherwise healthy people and has been reported more often in children and young adults. The fluid accumulation interferes with air movement into and out of the lungs, leading to shortness of breath, a persistent cough, chest tightness, and excessive fatigue. Disruption of normal breathing impairs blood oxygenation and, if it is severe enough, cyanosis (a bluish tint) of the lips and fingernails, mental confusion, and loss of consciousness may occur. High-altitude pulmonary edema is treated via

administration of supplemental oxygen and movement of the victim to a lower altitude.

High-Altitude Cerebral Edema

Rare cases of **high-altitude cerebral edema (HACE)**, which is fluid accumulation in the cranial cavity, have been reported. The condition is often a subsequent complication of HAPE. This neurological condition is characterized by mental confusion, lethargy, and ataxia (difficulty walking), progressing to unconsciousness and death. Most cases have been reported at altitudes greater than 4,300 m (14,108 ft). Similar to that for HAPE, the cause of HACE involves hypoxia-induced leakage of fluids from cerebral capillaries, causing edema and a resultant pressure buildup in the confined intracranial space. Treatment involves administration of supplemental oxygen, a hyperbaric bag, and prompt descent to a lower altitude. If descent is delayed, permanent impairment may ensue.

In Review

- Acute altitude sickness, often called acute mountain sickness, typically causes symptoms such as headaches, nausea, dyspnea, and insomnia. Symptoms usually appear 6 to 48 h after arrival at altitude.
 - The exact cause of acute altitude sickness is not known, but many researchers suspect that the symptoms may result from the combination of hypoxia and carbon dioxide accumulation in the tissues.
 - Acute altitude sickness usually can be avoided by a slow, gradual ascent to altitude, climbing no more than 300 m (984 ft) per day at elevations above 3,000 m (9,842 ft).
 - High-altitude pulmonary and cerebral edema (HAPE and HACE)—which involve accumulation of fluid in the lungs and cranial cavity, respectively—are life-threatening conditions. Both are treated by oxygen administration, hyperbaric bags, and descent.
-

IN CLOSING

Activities are seldom conducted under ideal environmental conditions. Heat, cold, humidity, and altitude—alone or in combination—present unique problems that are superimposed on the physiological demands of exercise. This chapter and the preceding chapter summarized the characteristics of these common environmental stresses and how we can cope with them.

Much of our discussion thus far has dealt with how physiological variables and environmental stress can hinder our performance. In the next part of the book, we examine various ways to optimize performance. We begin by looking at the importance of the amount of training, considering what happens when we train either too much or too little.

KEY TERMS

acute altitude (mountain) sickness

barometric pressure (P_b)

Cheyne-Stokes breathing

high-altitude cerebral edema (HACE)

high-altitude pulmonary edema (HAPE)

hypobaric

hypoxemia

hypoxia

partial pressure of oxygen (PO_2)

polycythemia

respiratory alkalosis

STUDY QUESTIONS

1. Describe the conditions at altitude that could limit the ability to perform physical activity.
2. What types of exercise are detrimentally influenced by exposure to high altitude and why?
3. When someone ascends to an altitude of over 1,500 m, describe the physiological adjustments that occur within the first 24 h.
4. Differentiate the physiological adjustments that accompany acclimation to altitude over a period of days, weeks, and months.
5. Would an endurance athlete who trained at altitude be able to perform better during subsequent sea-level performance? Why or why not?
6. Describe the theoretical advantage of living high and training low.

7. What are the best strategies for preparing athletes for high-altitude competition?
8. What are the health risks associated with acute exposure to high altitude and how can they be minimized?
9. How can the likelihood of experiencing acute altitude sickness be minimized? Once it occurs, how is it treated?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

[**www.HumanKinetics.com/PhysiologyOfSportAndExercise**](http://www.HumanKinetics.com/PhysiologyOfSportAndExercise)

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.